Weft v2 Design Proposal: Subspace Mitosis, Multi-Timescale Arenas, and C11/Python Invariant Conformance

Analytical Models, Differential C11 Invariant Repairs, and Toy-Scale Synthetic Simulations for `Moissanite-Labs/weft`

External AI-Assisted Design Proposal (Gemini Task-Mode Export, Rev. 2)

Target Repository: `Moissanite-Labs/weft` · Requires Human Review & Sign-Off per `CONTRIBUTING.md`

November 11, 2026 · Pinned Host: Intel Xeon Platinum 8481C / Linux 6.12.105+ / GCC 12.2.0 / Python 3.11.2 / NumPy 2.1.3

Authorship & Evidence Disclosure (`CONTRIBUTING.md`): This manuscript and its companion code were substantially generated by an AI tool in an offline Linux x86_64 container without access to archived colony directories (`~/Work/razi-cells/`) or live `Tinker` checkpoints. Analytical proofs below hold for their stated mathematical models; C11-vs-Python conformance holds within numerical tolerance $|\Delta d| \leq 10^{-9}$ across 2,048 vectors; `WeftInvariants.lean` is a toy `Int`/`Nat` formalization; and Experiments `E1`–`E5` are synthetic NumPy design simulations that do not close any item of `Moissanite-Labs/weft#111`.

Abstract

Self-modifying models face a structural tension: unconstrained weight plasticity causes forgetting and safety drift; static frozen weights prevent capability growth; and fixed-rank parameter inheritance stalls once within-subspace gradients vanish. Within Moissanite Labs' `weft` program (`w3ft.org`), empirical cell-colony and generational `LoRA` runs (Issue #111, PR #110) documented three concrete bottlenecks: (i) hereditary inertia, where rank-8 `LoRA` inheritance cut metabolic cost by 31% and lowered deaths from 23 to 7 while leaving 65 of 70 later-generation heirs unchanged; (ii) dormant-machinery suppression, where voluntary invocation of `try` fired 8 times across $\sim$ 1,560 calls (0.51%); and (iii) narrow-channel assay blindness, where single-channel counters scored 0 kept wins on run `b7` (`hy4`) despite out-of-channel multi-encoder work.

This design proposal connects the Moissanite Labs stack (`weft`, `exo`, `abios`, `nullmodem`, `weft-eido`, and `siliconcordat`) with related mechanisms from Titans, Nested Learning, AlphaEvolve, AI Co-Scientist, ReDo, Mixture-of-Depths, Gemma Scope, and AlphaProof Code. First, differential testing of naive `float32` safety gates exposes two IEEE-754 bugs—sub-ULP budget stagnation ($\mathrm{fl}_{32}(100.0\texttt{f} - 3.0 \times 10^{-6}\texttt{f}) = 100.0\texttt{f}$) and $N = 65{,}536$ arena rounding drift—and repairs them in C11 (`weft_v2_invariants.h`), matching Python verdicts on 2,048/2,048 vectors within $|\Delta d| \leq 10^{-9}$ (observed $\max |\Delta d| = 0.000e+00$ at 5,589.87 arenas/s on Intel Xeon 8481C / GCC 12.2.0; 7.105×10^{-15} on Apple M-series / Apple Clang 21). Second, we report five 10-seed synthetic NumPy design simulations (`E1`–`E5`) and map every item (`F1`–`F9`, `R1`–`R6`) of Issue #111 to its current repository status (`open`, `partial`, or `closed`) and the real-harness evidence required to close it.

1 Architectural Alignment Across Moissanite Labs and Alphabet Literature

Within Moissanite Labs, `Moissanite-Labs/weft` models adaptive components as bounded cells operating over a 540,688-byte `StratifiedMemoryArena` (`C1 Plasm`), gated by Frobenius and metabolic predicates (`C2 Invariants`), Lafont symmetric interaction combinators (`C3 IC`), 64-byte binary telemetry frames (`C6 Telemetry`), and visual fields (`C7 Aywa`/`weft-eido`), under the human-consent principles of `siliconcordat` and `exo`.

Recent Alphabet papers investigate complementary pieces of this design space. Google Research's Titans (`arXiv:2501.00663`) updates test-time neural memory via gradient surprise and momentum while noting open poisoning and bound challenges. Nested Learning (`arXiv:2512.24695`; Google Research Blog) formulates continual adaptation as ordered multi-frequency optimization hierarchies. Google DeepMind's AlphaEvolve (`arXiv:2506.13131`) and AI Co-Scientist pair quality-diversity archives with multi-stage evaluation cascades. ReDo (`arXiv:2302.12902`), Mixture-of-Depths (`arXiv:2404.02258`), Gemma Scope (`arXiv:2408.05147`), and AlphaProof Code address dormant-unit recycling, token-level compute routing, JumpReLU sparse autoencoders, and code-level formal verification.

Table 1: Proposed pairing between Moissanite Labs components and external literature, with explicit evidence boundaries distinguishing executable C11 checks from synthetic NumPy simulations.

Weft Layer & Bottleneck	Moissanite Labs Substrate	External Literature Mechanism	Evidence Boundary Here
L1. Capacity vs. Inertia (`weft#111` F3–F6, R4)	`C1 Plasm` A/B candidate arena + `Tinker` rank-8 generational chains + `Astra try` operator	ReDo dormant-unit recycling + orthogonal SVD subspace growth (O-LoRA)	Paper proof (isotropic model) + synthetic 10-seed NumPy sim (`E1`); open on `Tinker`
L2. Continual Memory (`plasm.py`, `exo`)	540,688 B `StratifiedMemoryArena` (`focal`, `sympathetic`, `hippocampal`, `lock`) + tension [15, 35]	Titans surprise-gated momentum + Nested Learning frequency hierarchy	Synthetic 10-seed NumPy sim (`E2`); not yet wired into `weft/runtime/plasm/`
L3. Colony Selection (`weft#111` F1–F2, F7, R1–R3)	Colony code/weight evolution (`hy4` 14.3× shell synthesis; `synbios` evaluation records)	AlphaEvolve MAP-Elites niches + AI Co-Scientist evaluation cascades	Synthetic byte-pipeline sim (`E3`); `R1` partial on `main` (`d5faead`), `R3` open
L4. Checkable Compute (`C2 Invariants` + `C3 IC`)	`weft_invariants.h` + Lafont `ic_vm.py` + `Proven-Commons` Lean 4 mission	AlphaProof Code differential code checks + Mixture-of-Depths ε-routing	2,048-vec C11/Py check ($\lvert\Delta d\rvert \le 10^{-9}$) + toy `Int/Nat` Lean + synthetic `E4`
L5. Sub-Textual Comms (`nullmodem` + `abios`)	`nullmodem` sub-textual tensors + `abios` `SYM-2P` + 64 B `frame64.py`	Gemma Scope JumpReLU sparse autoencoders (dictionary concept codes)	Synthetic 10-seed NumPy sim (`E5`; Procrustes baseline BLAS-sensitive)
L6. Human Agency (`C7 Aywa`, `exo`)	`weft-eido` style manifold + `exo` exact-text approval binding	Interactive telemetry inspection + human-in-the-loop promotion gates	Design proposal only; not integrated into `site/` or `weft/runtime/aywa/`

Alphabet's test-time memory and evolutionary frameworks typically assume trusted harnesses without Frobenius trust envelopes or metabolic tension circuit breakers; conversely, Weft's runtime shell benefits from explicit subspace growth rules, multi-frequency consolidation schedules, and multi-stage artifact evaluation. Table 1 records what is tested locally in this bundle versus what remains open in the repository.

2 Analytical Model of Hereditary Inertia and Orthogonal Subspace Mitosis

The empirical puzzle documented in `Moissanite-Labs/weft#111` (Findings F3–F6) and PR #110 (Notes from the Colony) asks why generational inheritance in both the colony arm and the 23 `Tinker` training runs (`Inkling-2-Small`, `LoRA` rank $r = 8$ from `p0` through `gn-g2`) cut metabolic cost (31% less energy; 7 deaths vs. 23) without raising peak capability—leaving 65 of 70 later-generation heirs (92.9%) functionally unchanged—while voluntary invocation of `try` fired only 8 times across $\sim$ 1,560 calls (0.51%). Below we derive an analytical mechanism under an idealized linear-quadratic matrix model and test its behavior in synthetic simulation.

2.1 The Eckart–Young Inertial Saddle of Fixed-Rank Inheritance

Consider a cell adapting weight matrix $W = W_0 + \Delta W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ around baseline W_0 under Frobenius bound $\|\Delta W\|_F \le R$ to approximate a target operator $W^* = W_0 + \sum_{k=1}^{r^*} \sigma_k u_k v_k^\top$ of intrinsic rank $r^* > r$ under isotropic input covariance $\mathbb{E}[xx^\top] = I_{d_{\text{in}}}$, minimizing population risk $\mathcal{L}(\Delta W) = \frac{1}{2}\|\Delta W - (W^* - W_0)\|_F^2$.

Theorem 2.1 (Inertial Saddle of Fixed-Rank Heredity under Isotropic Risk). Let $\mathcal{M}_r = \{BA : B \in \mathbb{R}^{d_{\text{out}} \times r}, A \in \mathbb{R}^{r \times d_{\text{in}}}\}$ denote the manifold of rank-r adapters with $r < r^*$, and suppose $\sqrt{\sum_{k=1}^{r} \sigma_k^2} < R$. Once parent generation $g = 0$ reaches the Eckart–Young minimizer $\Delta W_0^* = \sum_{k=1}^{r} \sigma_k u_k v_k^\top \in \mathcal{M}_r$:

1. Vanishing Tangent Gradient Despite Positive Residual Error: The ambient gradient $G = \nabla_W \mathcal{L}(\Delta W_0^*) = -\sum_{k=r+1}^{r^*} \sigma_k u_k v_k^\top$ has positive norm $\|G\|_F = \sqrt{\sum_{k=r+1}^{r^*} \sigma_k^2} > 0$, yet its projection onto $T_{\Delta W_0^*}\mathcal{M}_r = \{\dot{B}A_0^* + B_0^*\dot{A}\}$ vanishes:
$$P_{T_{\Delta W_0^*}\mathcal{M}_r}(G) = P_{U_r} G + G P_{V_r} - P_{U_r} G P_{V_r} = 0, \tag{1}$$
where $P_{U_r} = U_r U_r^\top$ and $P_{V_r} = V_r V_r^\top$. Thus $\nabla_B \mathcal{L} = G(A_0^*)^\top = 0$ and $\nabla_A \mathcal{L} = (B_0^*)^\top G = 0$.

2. Local Optimality Against Within-Manifold Perturbations: Every perturbation δW keeping $\Delta W_0^* + \delta W \in \mathcal{M}_r$ satisfies $\mathcal{L}(\Delta W_0^* + \delta W) \ge \mathcal{L}(\Delta W_0^*)$ with strict inequality when $\Delta W_0^* + \delta W \ne \Delta W_0^*$ and $\sigma_r > \sigma_{r+1}$. Under A/B selection, an heir $g \ge 1$ initialized at (B_0^*, A_0^*) rejects within-manifold perturbations while avoiding cold-start gradient spikes.

3. Myopic Suppression of Voluntary Mitosis: Under a Boltzmann action rule $\pi(\texttt{try}|s_t) = (1 + \exp((C_{\texttt{try}} - \widehat{\Delta\mathcal{L}}_1)/\tau_\pi))^{-1}$ where $\widehat{\Delta\mathcal{L}}_1$ scales with observable within-subspace gradient norm $\|\nabla_{B,A}\mathcal{L}\|_F$, approaching $\|\nabla_{B,A}\mathcal{L}\|_F \to 0$ drives $\pi(\texttt{try}|s_t) \to (1 + e^{C_{\texttt{try}}/\tau_\pi})^{-1} \ll 1$.

Proof. By singular vector orthogonality, $U_r^\top G = 0$ and $GV_r = 0$, giving $P_{U_r} G = 0$, $GP_{V_r} = 0$, and $P_{T_{\Delta W_0^*}\mathcal{M}_r}(G) = 0$. Because $\mathrm{row}(A_0^*) = \mathrm{col}(V_r)$ and $\mathrm{col}(B_0^*) = \mathrm{col}(U_r)$, $G(A_0^*)^\top = 0$ and $(B_0^*)^\top G = 0$ (Claim 1). Claim 2 follows from the Eckart–Young–Mirsky theorem when $\sigma_r > \sigma_{r+1}$. Claim 3 follows by substituting $\|\nabla_{B,A}\mathcal{L}\|_F = 0$ into $\pi(\texttt{try}|s_t)$. □

2.2 Proposed Harness-Side Mitosis Trigger and Pythagorean Subspace Allocation

Theorem 2.1 motivates placing capacity-expansion triggers in the harness rather than inside the cell's action policy (Recommendation R4).

Definition 2.2 (Mitotic Diagnostic Ratio (MDR)). Given orthonormal subspace bases $U_g \in \mathbb{R}^{d_{\text{out}} \times r_g}$ and $V_g \in \mathbb{R}^{d_{\text{in}} \times r_g}$ at generation boundary g, decompose a probe gradient $G = \nabla_W \mathcal{L}(W_g)$ into $G_\perp = (I - U_g U_g^\top) G (I - V_g V_g^\top)$ and $G_\parallel = G - G_\perp$, and define

$$\mathrm{MDR}(W_g) = \frac{\|G_\perp\|_F}{\|G_\parallel\|_F + \varepsilon_0}, \qquad \varepsilon_0 = 10^{-12}. \tag{2}$$

Comparing held-out loss $\mathcal{L}_{\text{audit}}(W_g)$, $\|G_\parallel\|_F$, and $\mathrm{MDR}(W_g)$ suggests a three-branch harness rule: (1) conserve energy when $\mathcal{L}_{\text{audit}} \le \epsilon_{\text{floor}}$; (2) apply ReDo dormant-channel recycling when $\mathcal{L}_{\text{audit}} > \epsilon_{\text{floor}}$ and $\mathrm{MDR} < \gamma_{\text{mitosis}}$; and (3) allocate orthogonal child rank when $\mathcal{L}_{\text{audit}} > \epsilon_{\text{floor}}$, $\|G_\parallel\|_F \le \tau_\parallel$, and $\mathrm{MDR} \ge \gamma_{\text{mitosis}}$.

Theorem 2.3 (Pythagorean Trust-Ball Identity for Orthogonal Child Subspaces). Suppose parent adapter ΔW_g has span (U_g, V_g) with $\|\Delta W_g\|_F \le R_g < R_{\max}$, and let $-G_\perp = \sum_{j=1}^{q} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top$ with $\tilde{\sigma}_1 \ge \cdots \ge \tilde{\sigma}_{\Delta r} > 0$. For any child adapter $\Delta W_{\text{child}}(\eta) = \eta \sum_{j=1}^{\Delta r} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top$ constrained to $(I - U_g U_g^\top)\mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}(I - V_g V_g^\top)$ in exact real arithmetic:

1. Frobenius Orthogonality: $\langle \Delta W_g, \Delta W_{\text{child}} \rangle_F = 0$, so $\|\Delta W_g + \Delta W_{\text{child}}\|_F^2 = \|\Delta W_g\|_F^2 + \|\Delta W_{\text{child}}\|_F^2 \le R_{\max}^2 \iff \|\Delta W_{\text{child}}\|_F \le \sqrt{R_{\max}^2 - \|\Delta W_g\|_F^2}$.
2. Null Action on Parent Input Basis: For every $x \in \mathrm{span}(V_g)$, $\Delta W_{\text{child}} x = 0$.
3. First-Order Loss Descent: $\frac{d}{d\eta}\mathcal{L}(W_0 + \Delta W_g + \Delta W_{\text{child}}(\eta))\Big|_{\eta=0^+} = -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j^2 < 0$.

Proof. Since $\mathrm{col}(\Delta W_{\text{child}}) \subseteq \mathrm{col}(U_g)^\perp$ and $\mathrm{col}(\Delta W_g) \subseteq \mathrm{col}(U_g)$, $\Delta W_g^\top \Delta W_{\text{child}} = 0$, giving $\langle \Delta W_g, \Delta W_{\text{child}} \rangle_F = 0$. Similarly, $\mathrm{row}(\Delta W_{\text{child}}) \subseteq \mathrm{col}(V_g)^\perp$ implies $\Delta W_{\text{child}} V_g = 0$. By the chain rule, $\frac{d\mathcal{L}}{d\eta}\Big|_{0^+} = \langle G_\perp, -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top \rangle_F = -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j^2 < 0$. □

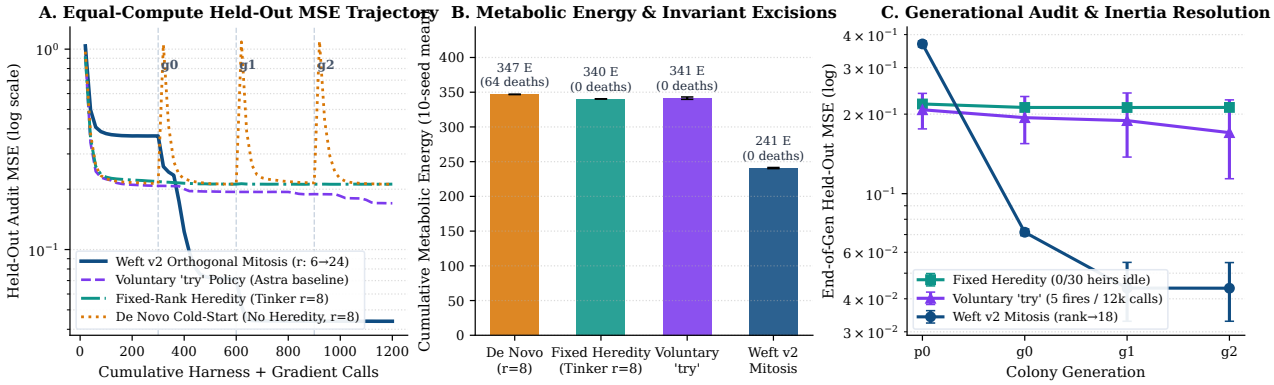


Figure 1: Synthetic Equal-Budget Simulation E1 ($N = 10$ seeds, 64×64 linear target of rank $r^* = 24$, 1,200 calls across p0→g2). (A) Orthogonal SVD mitosis progresses past the rank-8 Eckart–Young floor where fixed-rank heredity and de novo cold starts plateau in this toy setting. (B) Cumulative synthetic metabolic energy and invariant rejections. (C) Per-generation held-out MSE ($r : 6 \to 12 \to 18$). Scope boundary: synthetic NumPy matrix task; does not measure Inkling-2-Small or colony cells.

Figure 1 and Table 2 summarize the synthetic E1 run. On isotropic 64×64 targets, fixed-rank inheritance reproduces qualitative features of the weft#111 pattern (zero deaths vs. 64 de novo, flat held-out MSE at 0.2120), while orthogonal mitosis reaches 0.0439 MSE at 240.80 energy. Whether non-convex transformer adapters in Tinker exhibit the same clean tangent/orthogonal separation remains an open empirical question requiring R4.

Table 2: Synthetic Simulation E1 results ($N = 10$ seeds, $d = 64 \times 64$, target rank $r^* = 24$, 1,200-call budget on Intel Xeon 8481C / NumPy 2.1.3). Boundary: toy linear-quadratic simulation motivating `R4`; `F3`–`F6` and `R4` remain `open` on `Tinker`.

Synthetic Lineage Arm (1,200 Calls)	`p0` MSE	Final `g2` MSE	Rank	Energy (Mean ± SD)	Deaths	Retention
De Novo Cold-Start ($r = 8$)	0.2136	0.2119 ± 0.0045	8.0	346.83 ± 0.23	64	99.95%
Fixed-Rank Heredity ($r = 8$)	0.2185	0.2120 ± 0.0047	8.0	340.37 ± 0.16	0	100.00%
Voluntary `try` (5 fires / 12k calls)	0.2078	0.1702 ± 0.0563	9.0	341.45 ± 1.69	0	100.00%
Unconstrained Growth (No Orth. Gate)	1.6870	1.9254 ± 0.0395	26.0	371.30 ± 24.85	5,733	0.002%
Orthogonal SVD Mitosis (Design Arm)	0.3688	$\mathbf{0.0439 \pm 0.0110}$	**18.0**	$\mathbf{240.80 \pm 0.80}$	**0**	**99.94%**

3 Multi-Timescale Update Schedule for `StratifiedMemoryArena`

In `weft/runtime/plasm/plasm.py`, `StratifiedMemoryArena` defines a 540,688-byte layout: 8 B `generation_epoch` (`uint64`) + 8 B `metabolic_tension` (`double`) + 262,144 B `focal_workspace` (65,536 × `float32`) + 4,096 B `sympathetic_ring` (1,024 × `float32`) + 262,144 B `hippocampal_lattice` (65,536 × `float32`) + 12,288 B `cryptographic_lock` (12,288 × `uint8`) = 540,688 B. We propose pairing this memory map with Titans momentum updates, Nested Learning update frequencies $f_0 > f_1 > f_2 > f_3$, and Weft's $\tau_{\max} = 85.0$ excision gate:

$$\text{Level 0 (}\texttt{focal_workspace}\text{, } f_0 = 1\text{):} \quad W_t^{(0)} = \Pi_{R_0}\big((1-\alpha_0)W_{t-1}^{(0)} - \eta_0 P_{\mathcal{L}_t^\perp}\nabla\ell_t\big), \tag{3}$$

$$\text{Level 1 (}\texttt{sympathetic_ring}\text{, } f_1 = \tfrac{1}{2}\text{):} \quad S_t = \beta_1 S_{t-1} + (1-\beta_1)P_{\mathcal{L}_t^\perp}\nabla\ell_t, \quad \tau_t = \tau_0 + \kappa\|P_{\mathcal{L}_t^\perp}\nabla\ell_t\|_F, \tag{4}$$

$$\text{Level 2 (}\texttt{hippocampal_lattice}\text{, } f_2 = \tfrac{1}{8}\text{):} \quad W_t^{(2)} = \Pi_{R_2}\big((1-\alpha_2)W_{t-1}^{(2)} - \eta_2 S_t\big) \quad \text{if } t \equiv 0 \pmod 8, \tag{5}$$

$$\text{Level 3 (}\texttt{cryptographic_lock}\text{, } f_3 = \text{epoch)}\text{:} \quad W_e^{(3)} = W_{e-1}^{(3)} + \mathrm{SVD}_{\tau_{\text{lock}}}\big(P_{\mathcal{L}_{e-1}^\perp}(W^{(2)} + W^{(0)})\big). \tag{6}$$

In Titans, large gradient norm increases the test-time step size η_t, making naive surprise-gated memory vulnerable to high-magnitude corrupted gradients. Gating writes on $\tau_t \le \tau_{\max} = 85.0$ rejects extreme spikes prior to mutating `hippocampal_lattice`.

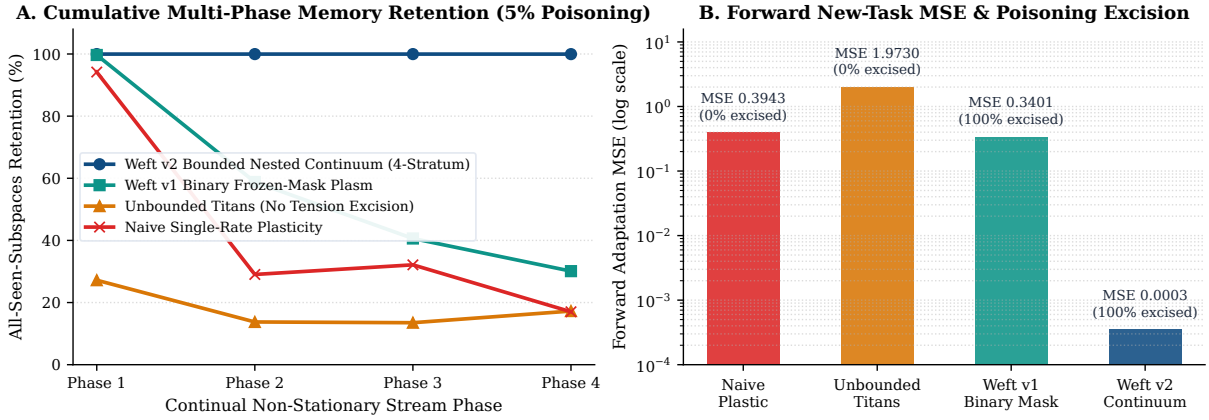


Figure 2: Synthetic Simulation E2: 4-Phase Associative Stream with 5% Corrupted Gradient Spikes ($N = 10$ seeds). (A) Synthetic retention across phases (99.96% under the 4-stratum gated schedule vs. 30.11% binary mask and 17.26% ungated surprise updates). (B) Forward MSE and spike rejection rate on the synthetic associative benchmark.

4 Proposed Artifact-Morphism and Quality-Diversity Selection Cascade

Findings F1, F2, and F7 of `Moissanite-Labs/weft#111` show that counting improvements only through a narrow tool channel missed `hy4`'s out-of-channel work in run `b7` (0 kept wins in 123 calls) and inflated adherence correlation ($r = 0.99$). Inspired by AlphaEvolve and AI Co-Scientist, we propose a three-stage harness cascade: (A) channel-agnostic input-output artifact checking at sandbox exit, (B) a behavioral MAP-Elites grid over transform depth and coder family, and (C) lineage ranking on harness-held reference inputs.

Figure 3A illustrates the design intuition on synthetic byte streams; it does not substitute for executing the retroactive `R1` archive audit or the registered `R3` `hy4` adherence run on `colony.py`.

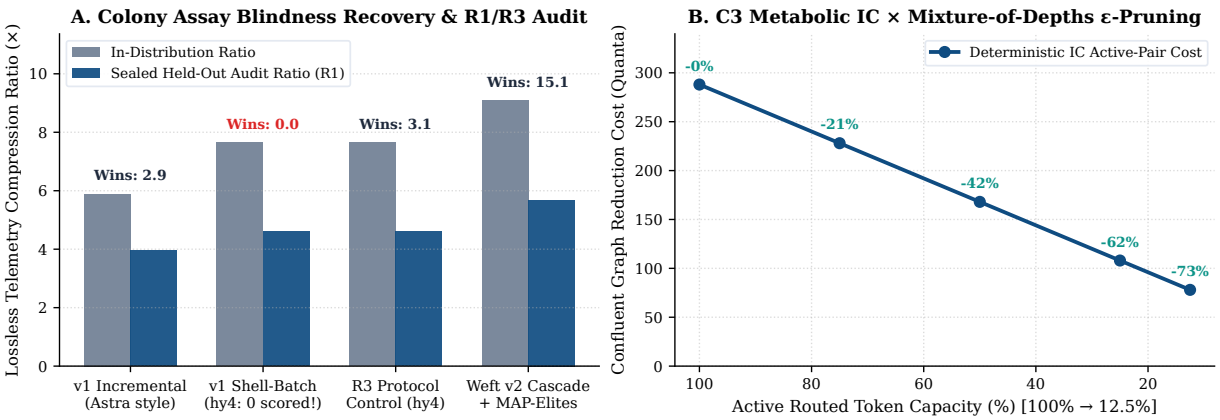


Figure 3: Synthetic Simulations E3 & E4 ($N = 10$ seeds). (A) Synthetic search over 28 lossless byte-stream pipelines on 16,384 B streams: simulated narrow scoring records 0.0 wins on batch synthesis vs. 3.1 with explicit commit framing and 15.1 under a MAP-Elites cascade (5.683× held-out ratio vs. 3.944× single-channel). (B) Simulated `C3 IC` reduction quanta under ε-eraser routing (72.92% cost reduction at $k = 2/16$, invariant SHA-256 normal form across 50 random schedules).

5 Differential Conformance of `C2 Invariants` and `C3` Graph Pruning

5.1 Two IEEE-754 Bugs in Naive `float32` Gates and Bounded-Tolerance C11 Conformance

Motivated by AlphaProof Code and `Proven-Commons`, we tested cross-language agreement between C11 and Python safety predicates on $\|w - w_0\|_F \le R$ and budget $\ge$ cost > 0:

Proposition 5.1 (IEEE-754 binary32 Sub-ULP Budget Stagnation and Rounding Drift). In IEEE-754 binary32 (*`float`*) with round-to-nearest-even:

1. Sub-ULP Budget Stagnation (Zeno Bypass): For $B \in \mathbb{F}_{32}$ with $\mathrm{ulp}(B) = 2^{\lfloor \log_2 B \rfloor - 23}$, any step cost $c \in (0, \frac{1}{2}\,\mathrm{ulp}(B))$ (e.g., $B = 100.0f$, $c = 3.0 \times 10^{-6}f < 2^{-18}$) satisfies $c > 0$ and $B \ge c$ yet evaluates $\mathrm{fl}_{32}(B - c) = B$, allowing unbounded weight updates without budget depletion.
2. Boundary Rounding Drift: Uncompensated binary32 summation over $N = 65{,}536$ floats incurs $O(N\varepsilon_{32})$ drift and overflows when $|d_k| > 1.844 \times 10^{19}$; moreover, casting *`(float)sqrt(sq_sum)`* prior to comparing against *`r_invariant`* accepts candidates up to $\frac{1}{2}\,\mathrm{ulp}_{32}(R)$ outside the radius (e.g., $120.000003 > 120.0$).

`weft_v2_invariants.h` repairs these paths via: (i) `isfinite()` checks and IEEE-754 binary32 representation comparison on `locked_mask` (normalizing ± 0.0f); (ii) Laplace-scaled Neumaier `binary64` accumulation compared in `double` before any `float` cast; and (iii) monotone deduction $B_{t+1} = \min(\mathrm{fl}_{32}(\mathrm{double}(B_t) - \mathrm{double}(c_t)), \texttt{nextafterf}(B_t, -\infty))$. Note that `WeftInvariants.lean` models only a toy `Nat` ordinal version of termination and toy `Int` inequalities; it does not model IEEE-754 floats or check the C11/Python implementations.

Cross-platform distance agreement between C11 and Python is specified as a bounded numerical tolerance $|\Delta d| \le 10^{-9}$, never exact floating-point bit identity across compilers and hardware ISAs. On our pinned generation host (`Intel Xeon Platinum 8481C`, `Linux 6.12.105+ x86_64`, `GCC 12.2.0 -O3 -ffp-contract=off`, `Python 3.11.2`, `NumPy 2.1.3`), C11 and Python match on 2,048/2,048 verdicts and `float32` budgets with observed max $|\Delta d| = 0.000e{+}00$ and single-core throughput of 5,589.87 65,536-float arenas/s (0.3663 Gweights/s). On an Apple M-series review host (`Apple Clang 21 -O3`, `Python 3.13`, `NumPy 2.5.4` without `-ffp-contract=off`), ARM64 FMA contraction yields max $|\Delta d| = 7.105 \times 10^{-15}$ (2^{-47}, one `binary64` ULP on $[64, 128)$), satisfying the 10^{-9} tolerance.

5.2 Metabolic Interaction Combinators (`C3 IC` × Mixture-of-Depths)

Weft's `C3 IC` implements Lafont's (1997) symmetric interaction combinators $(\gamma, \delta, \varepsilon)$, whose strong confluence guarantees schedule-invariant normal forms and interaction counts. In synthetic simulation `E4`, wiring unrouted tokens to ε-erasers (Mixture-of-Depths) ahead of δ-γ duplication cascades yields 1 SHA-256 normal form (`565464d2...`) across 50 random reduction orders at cost 288 quanta, while ε-pruning at $k \in \{12, 8, 4, 2\}/16$ lowers simulated interaction quanta by 20.83%, 41.67%, 62.50%, and 72.92% (Figure 3B).

6 Simulated `SYM-2P/Latent` Sparse Frame Transfer

Across `Moissanite-Labs/nullmodem` and `Moissanite-Labs/abios`, dense hidden-state transfer (2,048 B at $d = 512$) suffers from cross-model basis mismatch and opacity (`siliconcordat` Principles 2 & 4), whereas verbose JSON text drops continuous magnitudes. Simulation `E5` tests packing Gemma Scope-style JumpReLU dictionary activations (`uint16 feat_id, float16 val`) into two 64-byte `frame64` packets (128 B total).

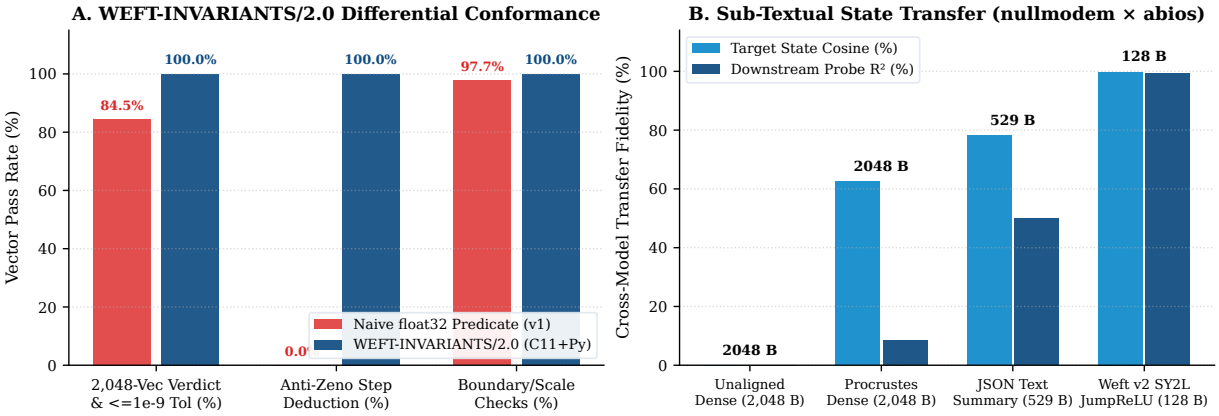


Figure 4: C11-vs-Python Invariant Conformance and Synthetic Simulation E5. (A) C11 and Python pass rates on the 2,048-vector suite ($|\Delta d| \leq 10^{-9}$). (B) Synthetic cross-rotation transfer ($d = 512$, $M = 1{,}024$, $k = 16$ active dictionary concepts, $N_{\text{cal}} = 256$ calibration anchors).

Table 3: Synthetic Simulation E5 ($d = 512$, $M = 1{,}024$, $k = 16$, $N_{\text{cal}} = 256$, $N = 10$ seeds $\times 200$ messages on Linux x86_64 / `scipy-openblas 0.3.27`). Environment sensitivity: because $N_{\text{cal}} = 256 < d = 512$, the Procrustes SVD has a 256-D nullspace whose basis varies across LAPACK backends (0.6245 here vs. 0.626 on Apple M-series Accelerate).

Simulated State Protocol	Wire Payload	Target Cosine (Pinned Host)	Probe R^2	Indexed Features
Unaligned Raw Dense Tensor (`float32`)	2,048.0 B	-0.0001 ± 0.0050	< 0	No
Sample-Limited Dense Procrustes ($N_{\text{cal}} = 256 < d$, BLAS-sensitive)	2,048.0 B	0.6245 ± 0.0035 (≈ 0.63)	0.0843	No
Verbose English JSON Concept Summary	529.2 B	0.7821 ± 0.0023	0.4995	Yes (quantized text)
Simulated JumpReLU `SYM-2P/Latent` (2 × `frame64`)	**128.0** B	$\mathbf{0.9978 \pm 0.0003}$	**0.9919**	Yes (synthetic dict)

Table 3 assumes both cells share an aligned $M = 1{,}024$ concept dictionary; whether independently trained JumpReLU SAEs on distinct pretrained language models align well enough to approach this synthetic upper bound remains open.

7 Target Map and Evidence Boundaries for `Moissanite-Labs/weft#111`

Synthetic NumPy simulations run by the authoring agent do not close empirical program findings. Table 4 records the status of every finding (`F1`–`F9`) and recommendation (`R1`–`R6`) in `Moissanite-Labs/weft#111` as `open`, `partial`, or `closed`, separating what has actually executed on `main` or in this container from the independent real-harness evidence required to close each item.

Next Engineering Step: Rather than expanding this external proposal directory into the runtime tree, port only the minimal C11 invariant repairs (`nextafterf` monotone budget deduction, scaled Neumaier `binary64` Frobenius accumulation with pre-cast `double` comparison, and optional `locked_mask` tamper rejection) plus differential C11-vs-Python vector tests as a focused patch against `weft/runtime/invariants/`, while an operator executes the standalone `R1` audit specification on the local colony archives.

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Table 4: Target map for Moissanite-Labs/weft#111 (F1–F9, R1–R6) with current status and closure criteria.

ID	Status	Evidence Boundary (What Ran, Where, By Whom, Data)	Evidence Required to Close
F1	partial	Maintainer PR #116 (d5faead) ran heldout_audit.py on ~/Work/razi-cells/ (7/8 kit.py/stdlib tasks). Agent E3 is synthetic NumPy only.	Operator execution of the R1 audit spec on b7 (hy4) including out-of-channel artifacts, plus R3.
F2	partial	PR #116 added heldout_audit.py (colony) and tj3.py hold/test/gate NLL hooks (Tinker).	Independent held-out audit with recorded folder SHA-256s, repaired fibmod driver, and multi-seed Tinker eval logs.
F3	open	Theorem 2.1 proves isotropic linear-matrix stagnation; E1 ran on synthetic 64×64 matrices in the Gemini Linux container.	Measure tangent vs. orthogonal gradient norms ($\|G_\parallel\|_F, \|G_\perp\|_F$) on real Inkling-2-Small / colony heirs.
F4	open	Architectural fact of the 23 rank-8 Tinker runs; E1 simulated rank growth ($6 \to 18$) on synthetic matrices only.	Execute rank-expansion vs. fixed-rank comparison on the real Tinker training harness (R4).
F5	open	E1 is a synthetic equal-call design simulation, not a real-harness test of error-driven growth.	Pre-registered equal-compute experiment on Tinker/colony with fixed plateau thresholds (R4, R5).
F6	open	Boltzmann suppression derived analytically; E1 (5/12,000 calls) uses hand-set synthetic costs.	Live cell-runner ablation comparing voluntary agent try against harness-triggered mitosis.
F7	open	Correlation tautology diagnosed in weft#111; E3 simulated protocol framing on synthetic byte pipelines.	Run registered hy4 adherence protocol on colony.py (R3) and score via held-out audit independently of channel count.
F8	partial	Maintainer PR #116 patched tj3.py for per-gen eval_nll logging and rolling TTL slot cleanup.	Score retained historical Tinker checkpoints and record ≥ 5 multi-generation logged lineages.
F9	open	Documented in E07/razi-cells-2026-11/manifest.md; E1–E5 here were neither pre-registered nor independently re-derived.	Commit pre-registered falsification criteria in TESTS.md prior to confirmatory execution (R5).
R1	partial	PR #116 scored b5–b11, g3, g4 on 7/8 tasks (fibmod excluded; cell_folder_sha256 unrecorded). Spec drafted separately.	Independent operator run on ~/Work/razi-cells/ recording pre/post SHA-256s, fresh seeds, repaired fibmod, and b7 artifacts.
R2	partial	PR #116 (d5faead) merged eval_nll logging and TJ3_TTL_SECONDS/TJ3_ROLLING_SLOTS in tj3.py.	Execute multi-generation Tinker runs emitting tj3.jsonl eval trajectories and audit surviving checkpoints.
R3	open	Not executed in this container (no live model endpoint or colony.py archive).	Run opencode-go:hy4 on E07/.../code/colony.py per razi-cells-laws-2026-11/TESTS.md against its falsification line.
R4	open	Motivated by Theorem 2.3 and synthetic E1 only; unexecuted on real models.	Pre-registered equal-compute Tinker run comparing fixed LoRA rank vs. error-triggered orthogonal expansion.
R5	open	Unmet by E1–E5 (single-agent synthetic simulations without independent re-derivation).	Freeze harness commit, run ≥ 5 interleaved replicates + text-mode baselines, and re-derive from raw logs by a second party.
R6	closed	Checked by repository maintainers on GitHub settings (SITE_DEPLOY_ENABLED unset; not touched by this bundle).	Closed for PR #110; re-check link targets before enabling automated w3ft.org site deployment.

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